DESIGN FOR ENGAGEMENT & IMPACT

Escape-Room-Style Learning Tied to Real Performance Tasks



PAMELA GUTMAN · DEVLEARN 2026

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POWERS AND ABILITIES

- **Learning Alchemy:** Transforms passive training into interactive adventures.
- **Quest Architecture:** Turns ordinary workplaces into Amazing Race-style missions, with AI as a creative sidekick.
- **Academic Side-Quest Completionist:** Two master's degrees, an Ed.D. in progress, and apparently no level cap.

KNOWN WEAKNESSES: Insatiable curiosity (both a blessing and a curse), pretty dice, excellent dark chocolate.



TODAY'S MISSION MAP

Play first. Then reverse-engineer what made it work.

01	PLAY	Solve a short team micro-escape
02	DECODE	Connect puzzle decisions to objectives
03	DESIGN	Mechanics, logistics, AI, and stakeholder buy-in
04	MAKE IT YOURS	Create a micro-escape; toolkit + Q&A


VANGUARD
HQ
EMERGENCY TRANSMISSION
THE
BACKUP
LEAGUE
Vanguard HQ needs you.

MISSION: THE BACKUP LEAGUE



01 Form a nearby team of about five.

02 Read the mission. Share evidence. Use hints.

03 One phone may help at the finale; a paper option exists.

2 MIN SETUP + 12 MIN PLAY

HINTS ARE IN YOUR PACKET, AND YOU CAN ASK ME FOR ADDITIONAL ASSISTANCE!


VANGUARD
HQ
MISSION COMPLETE
THE BACKUP LEAGUE
THE HEROES ARE FREE!

DEBRIEF

What happened at your team?

01 EVIDENCE

What information did you trust when clues conflicted?

02 DECISION

What choice did your group make, and how did you check it?

03 REVEAL

What did the completed pieces help you see together?

What you played. What you practiced.

THREE CHAPTERS + FINALE

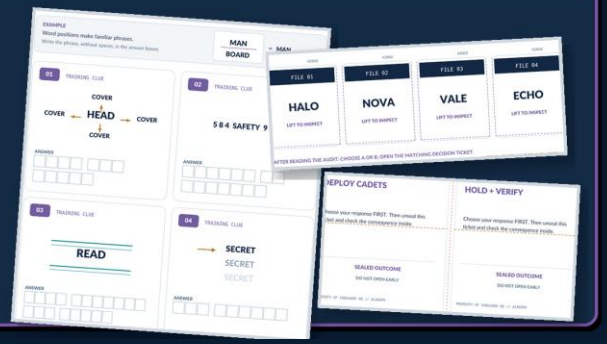
01 MEMORY MELTDOWN



02 GADGET GRAVEYARD



03 ACADEMY CONFIDENTIAL



04 CONSTELLATION FINALE



01 MISSION
Rescue the Vanguard without putting cadets at risk.

02 PLAYER DECISIONS
Prioritize dispatch, calibrate power, assess Academy risk.

03 FEEDBACK
Resolve chapters, earn star charts, unlock the console.

04 REAL-WORLD PRACTICE
Triage competing needs, apply constraints, verify decisions.

DESIGN FRAMEWORK

Performance-to-Puzzle Framework

PERFORMANCE GAP

01

What aren't people doing successfully, and why?

OBSERVABLE SUCCESS

02

What should learners be able to do, and what counts as success?

THEME + MECHANIC

03

What kind of puzzle will help, and how can we make it engaging?

AUTHENTIC CHALLENGE

04

What information, clues, rules, and materials will players need?

FEEDBACK + TRANSFER

05

How will they get help, know they're right, and apply the skill elsewhere?

START WITH PERFORMANCE. MAKE THE THINKING PLAYABLE.

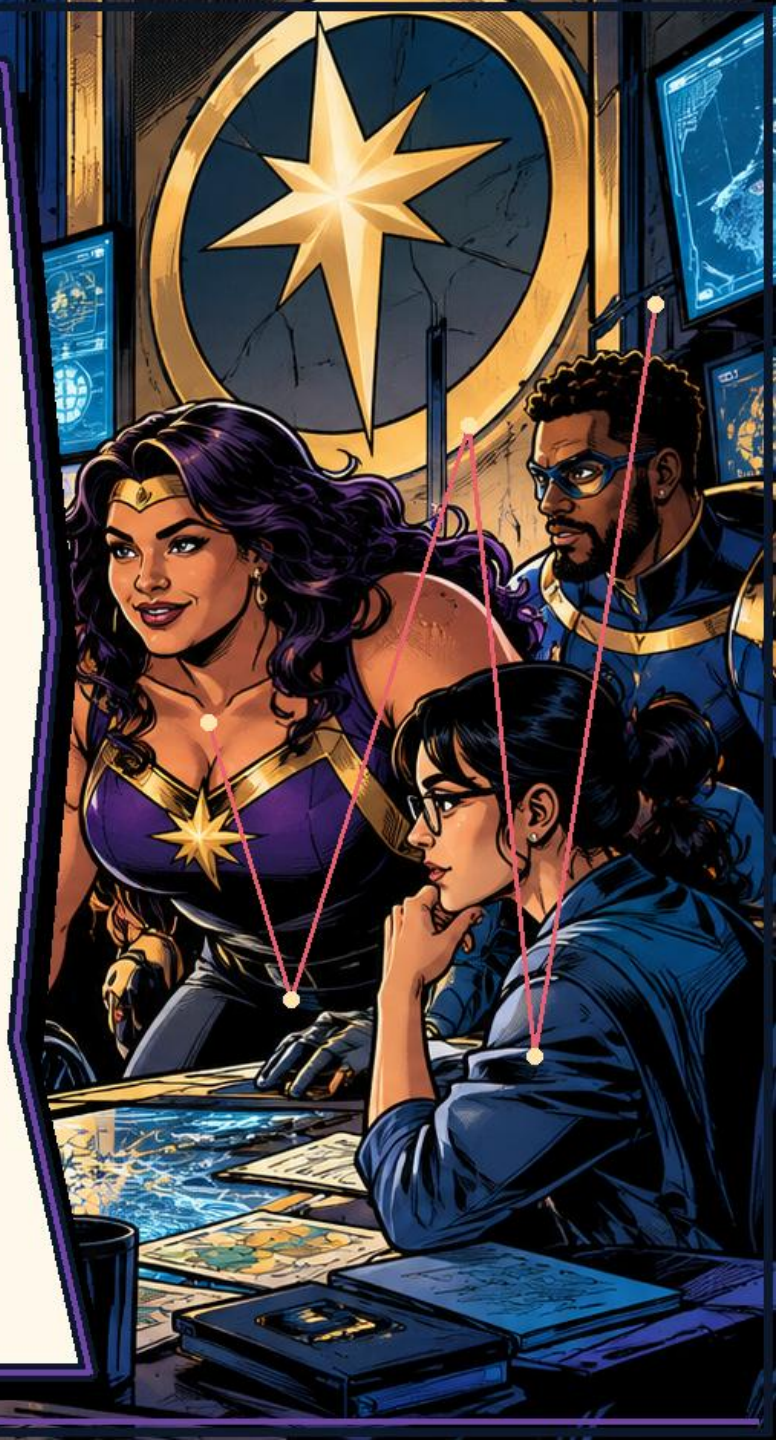
01 / FIND THE PERFORMANCE GAP

Find a behavior to improve before inventing a puzzle.

1	SPOT THE PATTERN	Where do people hesitate, make mistakes, or ask for help?
2	GATHER EVIDENCE	Observe work, review errors, or ask staff and supervisors.
3	CHECK THE CAUSE	Is this a skill gap, or a problem with tools or information?
4	DEFINE THE ACTION	State the decision people need to make more reliably.

RECAP: For this fictional demonstration, we chose a practice need; distinguishing urgent threats from distracting alerts.

YOUR OUTPUT · A specific practice-worthy behavior



02 / MAKE SUCCESS OBSERVABLE

Define a visible action and a clear standard for success.

GIVEN	DO	STANDARD	SCORE IT
What information or conditions will learners have?	Name an action you can see: sort, decide, explain.	What choices and evidence count as correct?	Could two observers agree on whether they succeeded?

RECAP: Given 6 incident reports, identify the highest priority ones and sequence them using the risk and timing criteria.

YOUR OUTPUT · An observable objective + success criterion





03 / CHOOSE MECHANIC + STORY

Choose the player actions first; then add the story.

SELECT MECHANIC(S)	ADD NARRATIVE	SKETCH THE FLOW
Choose an interaction that makes players do the kind of thinking that you want them to do on the job; sorting, sequencing, assembly, deduction, etc.	Wrap that action in an engaging fictional story; the players need a **role**, a **problem**, and a **reason** to solve it that matter to them.	Figure out the overall **sequence of events**; what will players need to know up front? What will they actually do? What happens when they succeed?

RECAP: Players select and sequence three incident cards, then flip them to verify the dispatch order. The superhero rescue supplies the stakes.

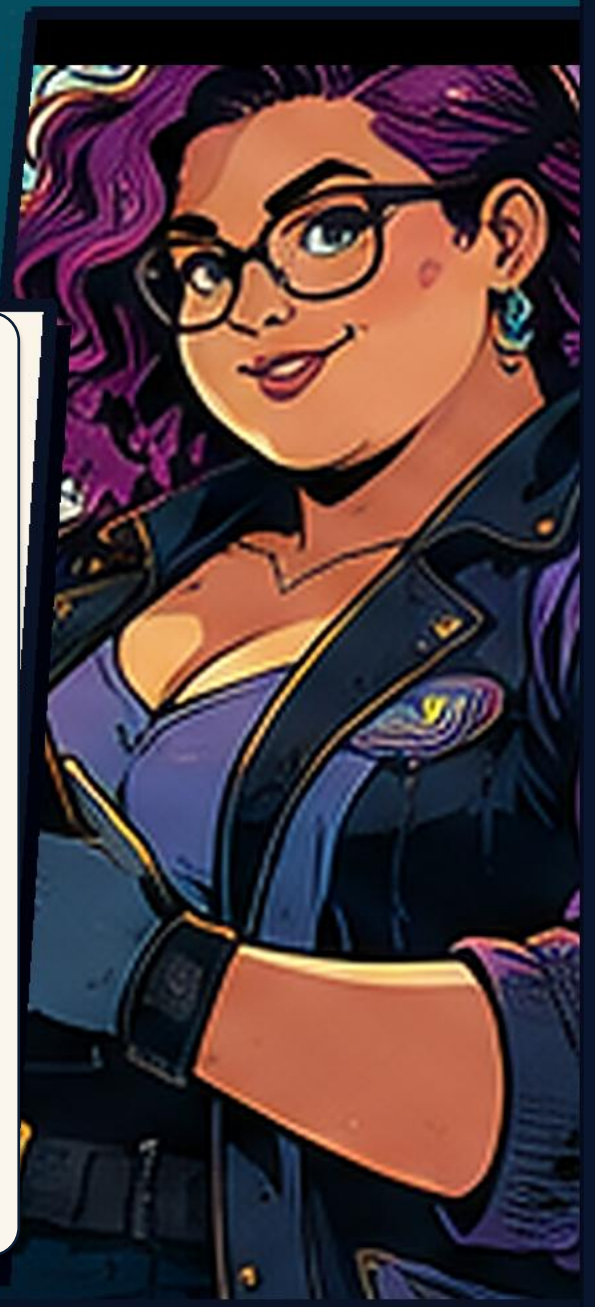
YOUR OUTPUT · A matched mechanic + meaningful story

PUZZLE TOOLBOX

Match the mechanic to the learning goal

LEARNERS NEED TO…	PUZZLE MECHANIC	DIY EXAMPLE
Prioritize risks	Sort / rank	Triage competing alerts
Follow a sequence	Order steps	Restore a process flow
Evaluate evidence	Deduction / logic	Reconcile conflicting reports
Apply constraints	Route / assemble	Configure safe equipment
Interpret information	Decode / overlay	Reveal meaningful data
Synthesize findings	Meta-puzzle	Combine chapter artifacts

ALIGNMENT TEST • Could they win without doing the target behavior?



PUZZLE ARCHITECTURE

Rachel Arpin: **Challenge → Action → Reward**

CHALLENGE

What is at stake?

ACTION

What must the players do?

REWARD

What changes when they succeed?

Each chapter has its own win following this framework.

The artifacts combine in the final meta-puzzle.





04 / BUILD THE CHALLENGE

Build clues and rules that require a defensible decision.

1	REQUIRED INFO	Identify what you need to make. What must players know to both understand the scenario and to make the correct decision?
2	CREATE ARTIFACTS	Time to make the materials! Generative AI can be a huge help here. Don't focus on making these pretty, they just need to work.
3	ESTABLISH SOLUTION	Once everything has been created, go through the puzzle yourself and verify the clues work as expected, and make an answer key.
4	PLAYTEST EVERYTHING	Testing identifies points of confusion, and player actions you may not have considered. Test with the target population if you can!

RECAP: 6 incident cards, three consequential threats, three plausible distractions, a recovery board, matching card backs, and an answer key.

YOUR OUTPUT · Playable artifacts + a verified solution (test, test, test!)

AI generated it. I had to make it work.

LOGIC-GRID CLUES

AI DRAFT: CLUES ALLOWED AMBIGUITY

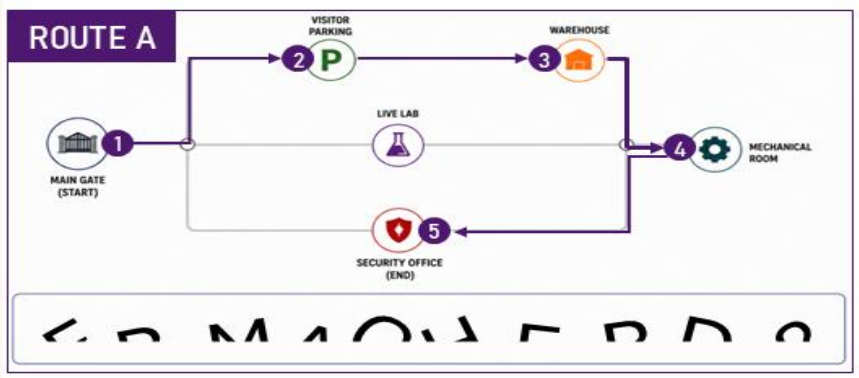
REVIEWERS	READINESS STATUS				REVIEW TIME			
	Awaiting Calibration	Under Repair	Ready for Test	Demo Ready	9:00	9:30	10:00	10:30
Jamie	✓	✗	✗	✗	✓	✗	✗	✗
Alex	✗	✓	✗	✗	✗	✗	✓	✗
Morgan	✗	✗	✗	✓	✗	✗	✗	✓
Taylor	✗	✗	✓	✗	✗	✓	✗	✗

Human revision

Revised the clues so the intended schedule could be uniquely deduced.

PATROL-ROUTE DRAWING

AI DRAFT: ROUTE BROKE A RULE



Human revision

Drew and checked every route manually in PowerPoint to ensure it was correct.

AI DRAFT → HUMAN VALIDATION → NOVICE PLAYTEST

EXAMPLE

Recap: From work problem to playable puzzle

1 PERFORMANCE GAP	2 OBSERVABLE SUCCESS	3 THEME + MECHANIC	4 AUTHENTIC CHALLENGE
Fictional HQ staff need to distinguish consequential threats from distractions	Select the three emergencies from six reports and rank them correctly	Sort and sequence cards to restore emergency dispatch	Create reports, distractors, recovery board, self-check, and verified solution

GO THROUGH THE PUZZLE WITH AN EYE ON HOW TRANSFERABLE THE SKILLS ARE TO THE REAL WORLD.

05 / DESIGN FEEDBACK + TRANSFER

Review feedback, identify hints, make changes, and plan for transfer.

DEBRIEF	Ask your playtesters to explain how they arrived at their conclusion and where they struggled along the way.
HELP	Identify where hints would help the players most, what they should be, and what to do if they run out of time.
UPDATE	Make any necessary changes to the materials, including self hints if using, and make the final versions.
BEYOND	If possible, work towards having players tackle a similar challenge to the puzzle on the job, to gauge transfer.

RECAP: Matching card backs verify the order; progressive hints support recovery; a new scenario would help evaluate transfer.

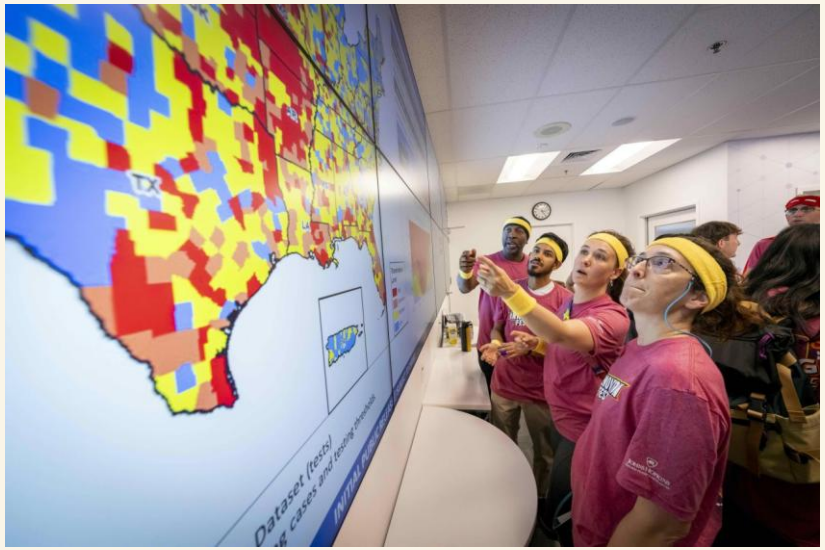
YOUR OUTPUT · Hint guide + updated final materials + transfer ideas



Case Study: APL's Amazing Race

A mash-up of scavenger hunt/escape-room-inspired activities; 10 teams of six staff members participated in physical challenges and puzzles that showcased staff members' quick thinking, APL-specific knowledge and overall resourcefulness.



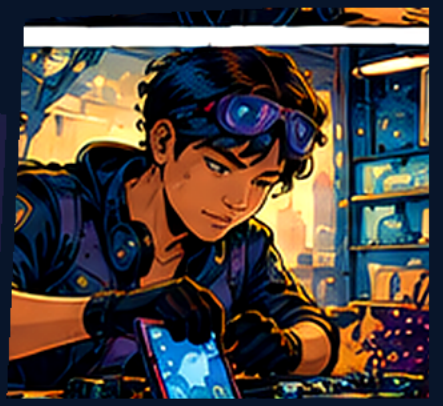




Lessons learned: problem → design response

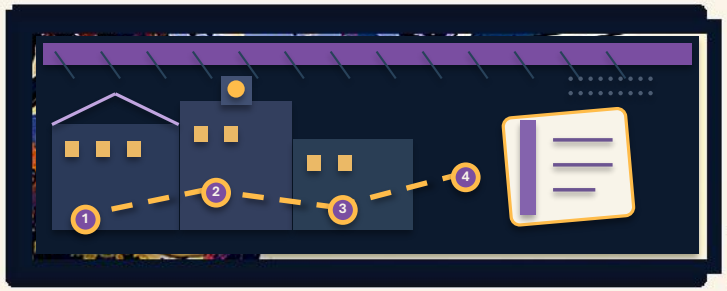
WHAT HAPPENED	WHAT TO DESIGN FOR
Teams stalled at hard stations	Clearer language in materials + tiered hints + a recovery path.
Routes and finish line confused teams	Updated wayfinding information + unmistakable completion.
Facilitators stretched; authors too familiar	Self-checks + new playtesters + easy reset with a Gamemaster guide.





One game architecture. Different learning goals.

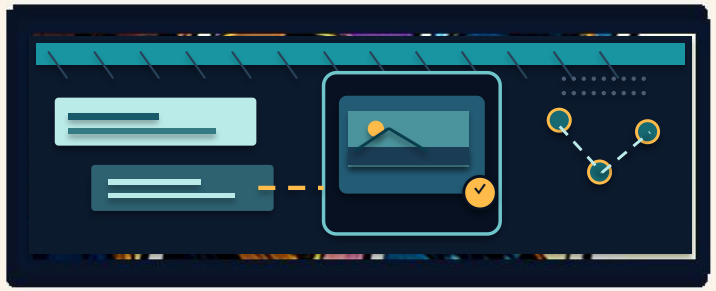
2023–24



LAB-WIDE

Innovation, teamwork, and campus discovery. Passport tracks clues.

2025



DISTRIBUTED + DISCOVERY

Slack clues + photo check-ins; first Discovery Program race.

2026



CROSS-COHORT

Cross-cohort navigation and shuttle practice. New puzzles; reused passport.

PRACTICAL LOGISTICS

Think carefully about the actual event space/people

EXAMPLE: 50 ATTENDEES / 10 TEAMS / 1 FACILITATOR

PACKETS

Prepacked kits, one per team; no station traffic.

NO TABLES

Rigid folder and large shared pieces.

SELF-CHECKS

Hints, answer validation, and recovery in every kit.

PACING

Time cues, optional bonus, shared ending for all.

BACKUPS

Paper QR alternative, spare kits, easy reset.

ACCESS

Readable print; no color-only clues; cognitive load in mind.

DRY RUN: packet distribution • access • sightlines • reset • tech fallback

STAKEHOLDER BUY-IN

Propose a small, measurable pilot

1 WHY THIS PROBLEM?

Can staff reliably identify and prioritize consequential incidents?

2 WHY THIS FORMAT?

The task requires comparing evidence and making decisions.

3 WHAT WILL IT TAKE?

Pilot the six-card dispatch challenge with a small group.

4 HOW WILL WE KNOW?

Check selections and reasoning; compare with a new scenario later.

PITCH: “Pilot [decision] with [practice task], then check [evidence].”

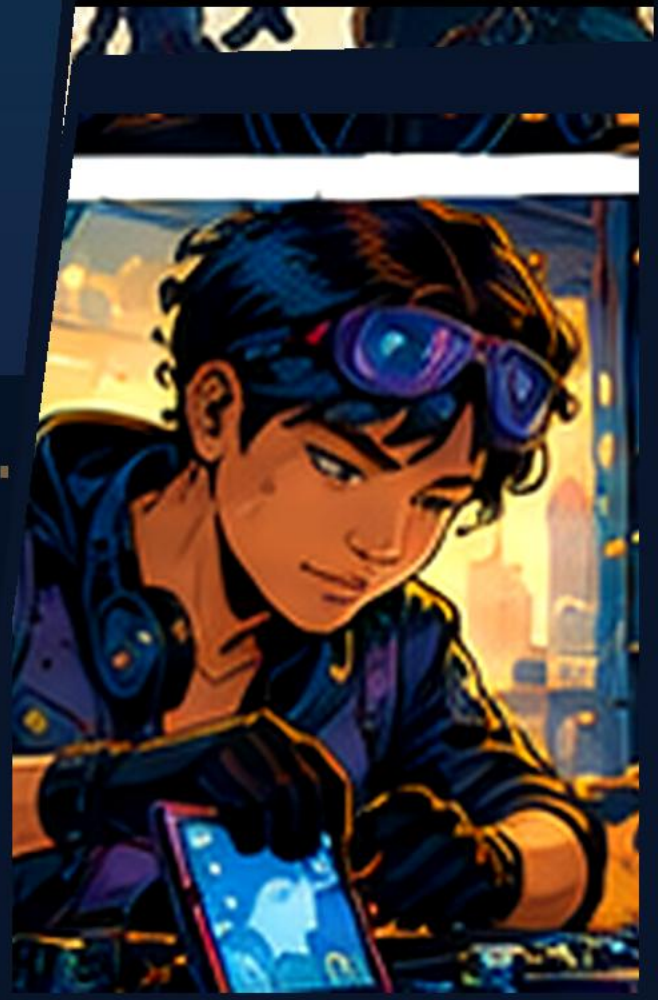
YOUR TURN

YOUR NEXT MISSION: DESIGN A MICRO-ESCAPE

TASK	SUCCESS	MECHANIC	FEEDBACK
What specific decision or behavior needs practice?	What does success look like?	What kind of interaction will get them there?	How will they get help, and know when they're right?

ALIGNMENT CHECK: Can someone win without using the skill?

60 SEC THINK · 90 SEC PAIR SHARE · 30 SEC VOLUNTEERS



TAKE THE TOOLKIT

One meaningful decision is enough to begin.

WHAT YOU CAN REUSE

01 **Field Guide + Five-Step Design Canvas**

02 **Puzzle Mechanics & Creative Reveals**

03 **Playtesting & Logistics Checklist**

04 **Pilot Proposal + AI Prompts**

RESOURCE SITE

https://tinyurl.com/DevLearnEscape



START SMALL · TEST EARLY · DEBRIEF WITH PURPOSE

FURTHER READING

Research and practical resources

DESIGN & RESEARCH

Rachel Arpin
Challenge–Action–Reward (ATD, 2023)

Veldkamp et al. (2020)
Escape education: systematic review

Fotaris & Mastoras (2022)
Room2Educ8: Design-thinking framework

PRACTICE & INSPIRATION

Rachel Arpin
Digital Escape Rooms for Leadership Training (2021)

CAST (2024)
UDL Guidelines 3.0

Constructed Adventures
DIY adventure ideas

GENERATIVE AI ATTRIBUTION

This presentation includes illustrations and graphic-novel backgrounds generated with ChatGPT (OpenAI, 2026) image-generation tools. ChatGPT was also used to assist with puzzle ideation and drafting. All materials were subject to human review and revision.